

17.21. Freeze Register

The freeze register can only be written once after an active reset and before motion starts. It is always readable.

FREEZE Register				
R/W	Addr	Bit	Val	Description
RW	0x4E	23:0	<i>DFREEZE</i> (Default: 0x000000)	
			U	Freeze event deceleration value. In case NFREEZE switches to low level, this parameter is used for an automatic linear ramp stop. Setting <i>DFREEZE</i> to 0 leads to an hard stop.
				Value representation: Frequency mode: not available Direct mode: [Δv per clk cycle] $a[\Delta v \text{ per clk_cycle}] = DFREEZE / 2^{37}$ $DFREEZE [\text{pps}^2] = DFREEZE / 2^{37} \cdot f_{CLK}^2$! Set <i>DFREEZE</i> ≤ 65536.
		31:24	<i>IFREEZE</i> (Default: 0x00)	
			U	Scaling value in case NFREEZE is tied low. In case <i>IFREEZE</i> =0, actual active scaling value is valid at FROZEN event.

Table 80: Freeze Register



17.22. Encoder Registers

Encoder Registers				
R/W	Addr	Bit	Val	Description
RW	0x50	31:0		<i>ENC_POS</i> (Default: 0x00000000)
			S	Actual encoder position [μsteps].
R				<i>ENC_LATCH</i> (Default: 0x00000000)
			S	Latched encoder position.
W	0x51	31:0		<i>ENC_RESET_VAL</i> (Default: 0x00000000)
			S	Defined reset value for <i>ENC_POS</i> in case the encoder position must be cleared to another value than 0.
R				<i>ENC_POS_DEV</i> (Default: 0x00000000)
			S	Deviation between <i>XACTUAL</i> and <i>ENC_POS</i> .
W	0x52	31:0		<i>CL_TR_TOLERANCE</i> (Default: 0x00000000) (Closed-loop operation)
			S	Tolerated absolute tolerance between <i>XACTUAL</i> and <i>ENC_POS</i> to trigger <i>TARGET_REACHED</i> (incl. <i>TARGET_REACHED_Flag</i> and event).
W	0x53	31:0		<i>ENC_POS_DEV_TOL</i> (Default: 0xFFFFFFFF)
			U	Maximum tolerated value of <i>ENC_POS_DEV</i> , which is not flagged as error.
W				<i>ENC_IN_RES</i> (Default: 0x00000000)
			U	Resolution [encoder steps per revolution] of the encoder connected to the encoder inputs.
R	0x54	30:0		<i>ENC_CONST</i> (Default: 0x00000000)
			U	Encoder constant. i Value representation: 15 digits and 16 decimal places
W		31		<i>manual_enc_const</i> (Default: 0)
				0 <i>ENC_CONST</i> will be calculated automatically. 1 Manual definition of <i>ENC_CONST</i> = <i>ENC_IN_RES</i>
W	0x55	31:0		<i>ENC_OUT_RES</i> (Default: 0x00000000)
			U	Resolution [encoder steps per revolution] of the serial encoder output interface.

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Encoder Registers				
R/W	Addr	Bit	Val	Description
W	0x56	15:0		<i>SER_CLK_IN_HIGH</i> (Default: 0x00A0)
			U	High voltage level time of serial clock output [# clock cycles].
		31:16		<i>SER_CLK_IN_LOW</i> (Default: 0x00A0)
			U	Low voltage level time of serial clock output [# clock cycles].
	0x57	15:0		<i>SSI_IN_CLK_DELAY</i> (Default: 0x0000)
			U	SSI encoder: Delay time [# clock cycles] between next data transfer after a rising edge of serial clock output. i In case <i>SSI_IN_CLK_DELAY</i> = 0: <i>SSI_IN_CLK_DELAY</i> = <i>SER_CLK_IN_HIGH</i>
				SPI encoder: Delay [# clock cycles] at start and end of data transfer between serial clock output and negated chip select. i In case <i>SSI_IN_CLK_DELAY</i> = 0: <i>SSI_IN_CLK_DELAY</i> = <i>SER_CLK_IN_HIGH</i>
	0x57	31:16		<i>SSI_IN_WTIME</i> (Default: 0x0F0)
			U	Delay parameter tw [# clock cycles] between two clock sequences for a multiple data transfer (of the same data). i SSI recommendation: tw < 19 μs.
	0x58	19:0		<i>SER_PTIME</i> (Default: 0x00190)
			U	SSI and SPI encoder: Delay time period tp [# clock cycles] between two consecutive clock sequences for new data request. i SSI recommendation: tp > 21 μs.

Table 81: Encoder Registers



17.23. PID & Closed-Loop Registers

PID and Closed-Loop Registers				
R/W	Addr	Bit	Val	Description
RW	0x59	31:0		<i>CL_OFFSET</i> (Default: 0x00000000) (Closed-loop operation)
			S	Offset between <i>ENC_POS</i> and <i>XACTUAL</i> after closed-loop calibration. It is set during closed-loop calibration process. It can be written manually.
W	0x5A	23:0		<i>PID_P</i> (Default: 0x0000000) (PID regulation)
U			Parameter P of PID regulator. Proportional term = $PID_E \cdot PID_P / 256$	
W				<i>CL_VMAX_CALC_P</i> (Default: 0x0000000) (Closed-loop operation)
U			Parameter P of PI regulator controls maximum catch-up velocity limitation.	
R		31:0		<i>PID_VEL</i> (Default: 0x00000000) (PID regulation)
			S	Actual PID output velocity.
W	0x5B	23:0		<i>PID_I</i> (Default: 0x0000000) (PID regulation)
U			Parameter I of PID regulator. Integral term = $PID_ISUM / 256 \cdot PID_I / 256$	
W				<i>CL_VMAX_CALC_I</i> (Default: 0x0000000) (Closed-loop operation)
U			Parameter I of PI regulator controls maximum catch-up velocity limitation.	
R		31:0		<i>PID_ISUM_RD</i> (Default: 0x00000000) (PID regulation)
			S	Actual PID integrator sum. Update frequency = $f_{CLK}/128$
W	0x5C	23:0		<i>PID_D</i> (Default: 0x0000000) (PID regulation)
U			Parameter D of PID regulator. PID_E is sampled with $f_{CLK} / 128 / PID_D_CLKDIV$. Derivative term = $(PID_E_{LAST} - PID_E_{ACTUAL}) \cdot PID_D$	
W				<i>CL_DELTA_P</i> (Default: 0x0000000) (Closed-loop operation)
U			Gain parameter that is multiplied with the actual position difference in order to calculate the actual commutation angle for position maintenance stiffness. Clipped at <i>CL_BETA</i> . Real value = $CL_DELTA_P / 2^{16}$; Ex: 65536 → 1.0 (gain=1) Value representation: 8 digits and 16 decimal places.	
W	0x5D	14:0		<i>PID_I_CLIP</i> (Default: 0x0000) (PID regulation) (Closed-loop operation)
U			Clipping parameter for <i>PID_ISUM</i> . Real value = $PID_ISUM \cdot 2^{16} \cdot PID_ICLIP$	
W		23:16		<i>PID_D_CLKDIV</i> (Default: 0x00) (PID regulation)
U			Clock divider for D part calculation.	
R		31:0		<i>PID_E</i> (Default: 0x00000000) (PID regulation)
			S	Actual position deviation.
W	0x5E	30:0		<i>PID_DV_CLIP</i> (Default: 0x00000000) (PID regulation) (Closed-loop operation)
			U	Clipping parameter for <i>PID_VEL</i> .
W	0x5F	19:0		<i>PID_TOLERANCE</i> (Default: 0x00000) (PID regulation)
U			Tolerated position deviation: $PID_E = 0$ in case $ PID_E < PID_TOLERANCE$	
W		7:0		<i>CL_TOLERANCE</i> (Default: 0x00) (Closed-loop operation)
			U	Tolerated position deviation: $CL_DELTA_P = 65536$ (gain=1) in case $ ENC_POS_DEV < CL_TOLERANCE$



Table 82: PID and Closed-Loop Registers

17.24. Miscellaneous Registers

Miscellaneous Registers				
R/W	Addr	Bit	Val	Description
W	0x60	23:0		<i>FS_VEL</i> (Default: 0x000000) (Open-loop operation)
			U	Minimum fullstep velocity [pps]. In case $ V_{ACTUAL} > FS_VEL$ fullstep operation is active, if enabled.
				<i>DC_VEL</i> (Default: 0x000000) (dcStep only)
			U	Minimum dcStep velocity [pps]. In case $ V_{ACTUAL} > DC_VEL$ dcStep is active, if enabled.
				<i>CL_VMIN_EMF</i> (Default: 0x000000) (Closed-loop operation)
			U	Encoder velocity at which back-EMF compensation starts.
	0x61	7:0		<i>DC_TIME</i> (Default: 0x00) (TMC26x only and dcStep only)
			U	Upper PWM on-time limit for commutation. i Set slightly above effective blank time TBL of the driver.
		15:8		<i>DC_SG</i> (Default: 0x0000) (TMC26x and dcStep only)
			U	Maximum PWM on-time [# clock cycles · 16] for step loss detection. If a loss is detected (step length of first regular step after blank time of the dcStep input signal is below <i>DC_SG</i>), a stall event will be released.
		31:16		<i>DC_BLKTIME</i> (Default: 0x0000) (TMC26x and dcStep only)
			U	Blank time [# clock cycles] after fullstep release when no signal comparison should happen.
	0x62	23:0		<i>CL_VADD_EMF</i> (Default: 0x000000) (Closed-loop operation)
			U	Additional velocity value to calculate the encoder velocity at which back-EMF compensation reaches the maximum angle <i>CL_GAMMA</i> .
		31:0		<i>DC_LSPTM</i> (Default: 0x00FFFFFF) (dcStep only)
			U	dcStep low speed timer [# clock cycles]
	0x62	23:0		<i>ENC_VEL_ZERO</i> (Default: 0xFFFFF) (dcStep is not enabled)
			U	Delay time [# clock cycles] after the last incremental encoder change to set $V_{ENC_MEAN} = 0$.

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Miscellaneous Registers				
R/W	Addr	Bit	Val	Description
W	0x63	7:0		<i>ENC_VMEAN_WAIT</i> (Default:0x00) (incremental encoders only)
			U	Delay period [# clock cycles] between two consecutive actual encoder velocity values that account for calculation of mean encoder velocity. ! Set <i>ENC_VMEAN_WAIT</i> > 32. i Is set automatically to <i>SER_PTIME</i> for absolute SSI/SPI encoder.
		7:0		<i>SER_ENC_VARIATION</i> (Default:0x00) (absolute encoders only)
			U	Multiplier for maximum permitted serial encoder variation between consecutive absolute encoder requests. ! Maximum permitted value = $ENC_VARIATION / 256 \cdot 1/8 \cdot ENC_IN_RES$. ! If <i>ENC_VARIATION</i> = 0: Maximum permitted value = $1/8 \cdot ENC_IN_RES$.
		11:8		<i>ENC_VMEAN_FILTER</i> (Default:0x0)
			U	Filter exponent to calculate mean encoder velocity.
		31:16		<i>ENC_VMEAN_INT</i> (Default:0x0000) (incremental encoders only)
			U	Encoder velocity update time [# clock cycles]. i Minimum value is set automatically to 256.
W	0x64	31:0		<i>CL_CYCLE</i> (Default:0x0000) (absolute encoders only)
			U	Closed-loop control cycle [# clock cycles]. i Is set automatically to <i>fastest</i> possible cycle for ABN encoders.
W	0x64	31:0		Reserved. Set to 0x00000000.
R	0x65	31:0		<i>V_ENC</i> (Default:0x00000000)
			U	Actual encoder velocity [pps].
R	0x66	31:0		<i>V_ENC_MEAN</i> (Default:0x00000000)
			S	Filtered encoder velocity [pps].
W	0x67	23:0		<i>VSTALL_LIMIT</i> (Default:0x00000000)
			U	Stop on stall velocity limit [pps]: Only above this limit an active stall leads to a stop on stall, if enabled.
W	0x7C	31:0		<i>CIRCULAR_DEC</i> (Default:0x000)
			U	Decimal places for circular motion if one revolution is not exactly mapped to an even number of μ Steps per revolution. Value representation: 1 digit and 31 decimal places.
W	0x7D	15:0		<i>ENC_COMP_XOFFSET</i> (Default:0x0000)
			U	Start offset for triangular compensation in horizontal direction. $0 \leq ENC_COMP_XOFFSET < 2^{16}$
		23:16		<i>ENC_COMP_YOFFSET</i> (Default:0x00)
			S	Start offset for triangular compensation in vertical direction. $-128 \leq ENC_COMP_YOFFSET \leq 127$
W	0x7E	31:24		<i>ENC_COMP_AMPL</i> (Default:0x00)
			U	Maximum amplitude for encoder compensation.

Table 83: Miscellaneous Registers



17.25. Transfer Registers

Transfer Registers				
R/W	Addr	Bit	Val	Description
W	0x68	31:0		<i>ADDR_TO_ENC</i> (Default:0x00000000) (SPI encoders only)
			-	Address data permanently sent to get encoder angle data from the SPI encoder slave device.
			-	Address data sent from TMC4361 to SPI encoder for one-time data transfer.
W	0x69	31:0		<i>DATA_TO_ENC</i> (Default:0x00000000) (SPI encoders only)
			-	Configuration data sent from TMC4361 to SPI encoder for one-time data transfer.
R	0x6A	31:0		<i>ADDR_FROM_ENC</i> (Default:0x00000000) (SPI encoders only)
			-	Repeated request data is stored here.
			-	Address data received from SPI encoder as response of the one-time data transfer.
R	0x6B	31:0		<i>DATA_FROM_ENC</i> (Default:0x00000000) (SPI encoders only)
			-	Data received from SPI encoder as response of the one-time data transfer.
W	0x6C	31:0		<i>COVER_LOW</i> (Default:0x00000000)
			-	Lower configuration bits of SPI orders that can be sent from TMC4361 to the motor drivers via SPI output.
			-	Automatic cover data transfer (<i>automatic_cover</i> = 1): Value in <i>COVER_LOW</i> are sent in case <i>VACTUAL</i> crosses <i>SPI_SWITCH_VEL</i> downwards. ! Set <i>COVER_DATA_LENGTH</i> ≤ 32. ! In case <i>COVER_DATA_LENGTH</i> = 0, no TMC2130 must be selected.
W	0x6D	31:0		<i>COVER_HIGH</i> (Default:0x00000000)
			-	Upper configuration bits of SPI orders that can be sent from TMC4361 to the motor drivers via SPI output.
			-	Automatic cover data transfer (<i>automatic_cover</i> = 1): Value in <i>COVER_LOW</i> are sent if <i>VACTUAL</i> crosses <i>SPI_SWITCH_VEL</i> upwards. ! Set <i>COVER_DATA_LENGTH</i> ≤ 32. ! In case <i>COVER_DATA_LENGTH</i> = 0, no TMC2130 must be selected.
R	0x6E	31:0		<i>COVER_DRV_LOW</i> (Default:0x00000000)
			-	Lower configuration bits of SPI response received from the motor driver connected to the SPI output.
R	0x6F	31:0		<i>COVER_DRV_HIGH</i> (Default:0x00000000)
			-	Upper configuration bits of SPI response received from the motor driver connected to the SPI output.

Table 84: Transfer Registers



17.26. SinLUT Registers

SinLUT Registers				
R/W	Addr	Bit	Val	Description
W	0x70	31:0		<i>MSLUT[0]</i> (Default:0xAAAAB554)
	0x71			<i>MSLUT[1]</i> (Default:0x4A9554AA)
	0x72			<i>MSLUT[2]</i> (Default:0x24492929)
	0x73			<i>MSLUT[3]</i> (Default:0x10104222)
	0x74			<i>MSLUT[4]</i> (Default:0xFBFFFFFF)
	0x75			<i>MSLUT[5]</i> (Default:0xB5BB777D)
	0x76			<i>MSLUT[6]</i> (Default:0x49295556)
	0x77			<i>MSLUT[7]</i> (Default:0x00404222)
			-	! Each bit defines the difference between consecutive values in the microstep look-up table MSLUT (in combination with <i>MSLUTSEL</i>).
W	0x78	31:0		<i>MSLUTSEL</i> (Default:0xFFFF8056)
			-	Definition of the four segments within each quarter MSLUT wave.
R	0x79	9:0		<i>MSCNT</i> (Default:0x000)
			U	Actual μ Step position of the sine value.
W				<i>MSOFFSET</i> (Default:0x000) (TMC23x/24x only)
			U	Microstep offset for PWM mode.
R	0x7A	8:0		<i>CURRENTA</i> (Default:0x000)
			S	Actual current value of coilA (sine values).
		24:16		<i>CURRENTB</i> (Default:0x000)
			S	Actual current value of coilB (sine90_120 values).
R	0x7B	8:0		<i>CURRENTA_SPI</i> (Default:0x000)
			S	Actual scaled current value of coilA (sine values) that are sent to the motor driver.
		24:16		<i>CURRENTB_SPI</i> (Default:0x000)
			S	Actual scaled current value of coilB (sine90_120 values) that are sent to the motor driver.
R	0x7C	8:0		<i>SCALE_PARAM</i> (Default:0x000)
			U	Actual used scale parameter.
W	0x7E	7:0		<i>START_SIN</i> (Default:0x00)
			U	Start value for sine waveform.
		31:16		<i>START_SIN90_120</i> (Default:0xF7)
			U	Start value for cosine waveform.
		31:24		<i>DAC_OFFSET</i> (Default:0x00)
			U	Offset for absolute sine and cosine values, which are forwarded via SPI output as DAC output values.

Table 85: SinLUT Registers



17.27. TMC Version Register

Version Register				
R/W	Addr	Bit	Val	Description
R	0x7F	15:0		Version No (Default:0x0001)
			U	TMC4361 version number.

Table 86: Vision Registers



18. Absolute Maximum Ratings

The maximum ratings may not be exceeded under any circumstances. Operating the circuit at or near more than one maximum rating at a time for extended periods shall be avoided by application design.

Maximum Ratings: 3.3V supply				
Parameter (VCC = 3.3V nominal → TEST_MODE = 0V)	Symbol	Min	Max	Unit
Supply voltage	V _{CC}	3.0	3.6	V
Input voltage IO	V _{IN}	−0.3	3.6	V

Table 87: Maximum Ratings: 3.3V supply

Maximum Ratings: 5.0V supply				
Parameter (VCC = 5V nominal → TEST_MODE = 1.8V)	Symbol	Min	Max	Unit
Supply voltage	V _{CC}	4.8	5.2	V
Input voltage IO	V _{IN}	−0.3	5.2	V

Table 88: Maximum Ratings: 5.0V supply

Maximum Ratings: Temperature				
Parameter	Symbol	Min	Max	Unit
Temperature	T	−40	125	°C

Table 89: Maximum Ratings: Temperature



19. Electrical Characteristics

DC characteristics contain the spread of values guaranteed within the specified supply voltage range unless otherwise specified. Typical values represent the average value of all parts measured at +25°C. Temperature variation also causes stray to some values. A device with typical values will not leave Min/Max range within the full temperature range.

DC Characteristics						
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Extended temperature range	T_{COM}		-40°C		125	°C
Nominal core voltage	V_{DD}			1.8		V
Nominal IO voltage	V_{DD}			3.3 / 5.0		V
Nominal input voltage	V_{IN}		0.0		3.3 / 5.0	V
Input voltage low level	V_{INL}	$V_{DD} = 3.3V / 5V$	-0.3		0.8 / 1.2	V
Input voltage high level	V_{INH}	$V_{DD} = 3.3V / 5V$	2.3 / 3.5		3.6 / 5.2	V
Input with pull-down		$V_{IN} = V_{DD}$	5	30	110	μA
Input with pull-up		$V_{IN} = 0V$	-110	-30	-5	μA
Input low current		$V_{IN} = 0V$	-10		10	μA
Input high current		$V_{IN} = V_{DD}$	-10		10	μA
Output voltage low level	V_{OUTL}	$V_{DD} = 3.3V / 5V$			0.4	V
Output voltage high level	V_{OUTH}	$V_{DD} = 3.3V / 5V$	2.64 / 4.0			V
Output driver strength	I_{OUT_DRV}	$V_{DD} = 3.3V / 5V$		4.0		mA

Table 90: DC Characteristics

19.1. Power Dissipation

Power Dissipation						
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Static power dissipation	PD_{STAT}	All inputs at VDD or GND $V_{DD} = 3.3V / 5V$			1.1 / 1.7	mW
Dynamic power dissipation	PD_{DYN}	All inputs at VDD or GND f_{CLK} variable $V_{DD} = 3.3V / 5V$			2.3 / 3.7	mW / MHz
Total power dissipation	PD	$f_{CLK} = 16\text{ MHz}$ $V_{DD} = 3.3V / 5V$			37.9 / 60.3	mW

Table 91: Power Dissipation



19.2. General IO Timing Parameters

General IO Timing Parameters						
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Operation frequency	f_{CLK}	$f_{\text{CLK}} = 1 / t_{\text{CLK}}$	4.2 ¹⁾	16	30	MHz
Clock Period	t_{CLK}	Rising edge to rising edge	33.5	62.5		ns
Clock time low			16.5			ns
Clock time high			16.5			ns
CLK input signal rise time	$t_{\text{RISE_IN}}$	20 % to 80 %			20	ns
CLK input signal fall time	$t_{\text{FALL_IN}}$	80 % to 20 %			20	ns
Output signal rise time	$t_{\text{RISE_OUT}}$	20 % to 80 % load 32 pF		3.5		ns
Output signal fall time	$t_{\text{FALL_OUT}}$	80 % to 20 % load 32 pF		3.5		ns
Setup time for SPI input signals in synchronous design	t_{SU}	Relative to rising clk edge	5			ns
Hold time	t_{HD}	Relative to rising clk edge	5			ns

Table 92: General IO Timing Parameters

¹⁾ The lower limit for f_{CLK} refers to the limits of the internal unit conversion to physical units. The chip will also operate at lower frequencies.



19.3. Layout Examples

19.3.1. Internal Circuit Diagram for Layout Example

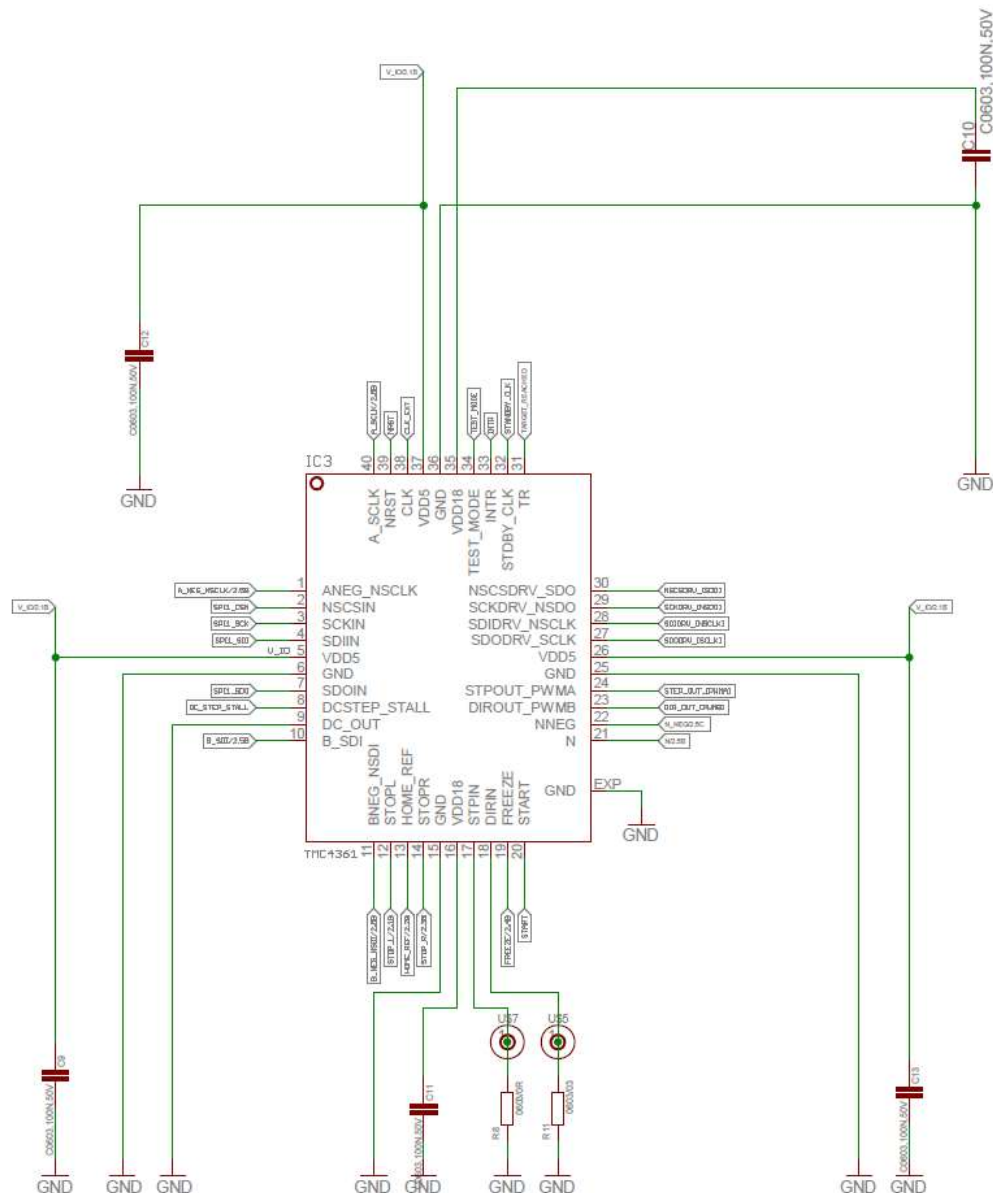


Figure 67: Internal Circuit Diagram for Layout Example



19.3.2. Components Assembly for Application with Encoder

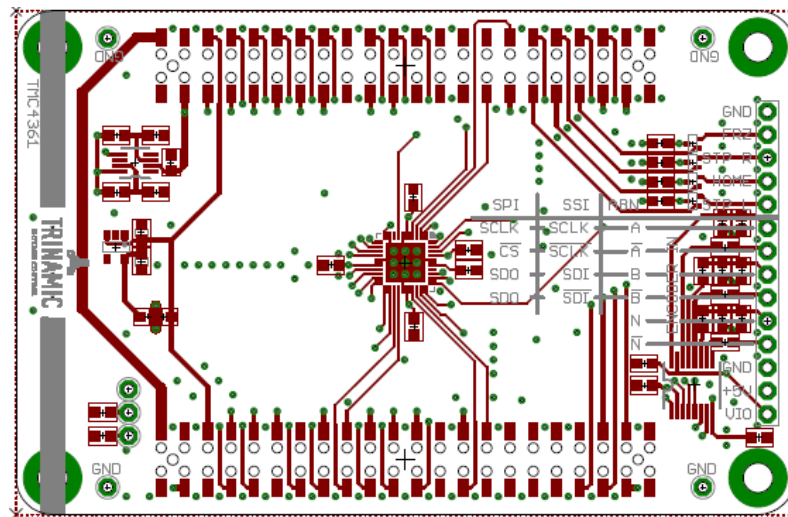


Figure 68: Components Assembly for Application with Encoder

19.3.3. Top Layer: Assembly Side

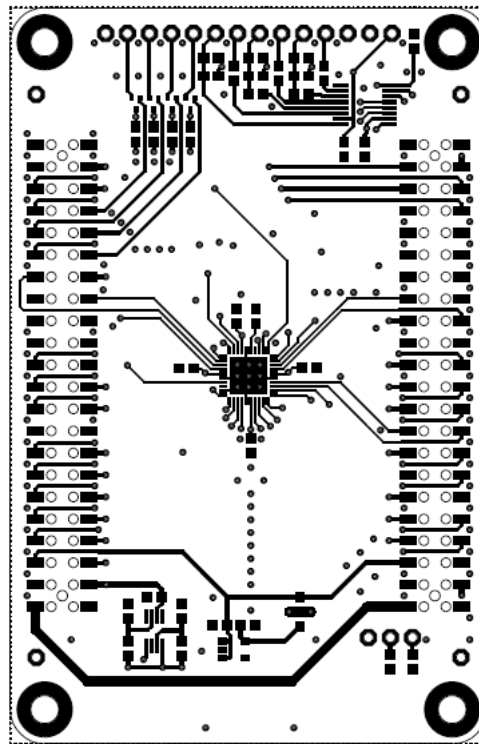


Figure 69: Top Layer: Assembly Side



19.3.4. Inner Layer (GND)

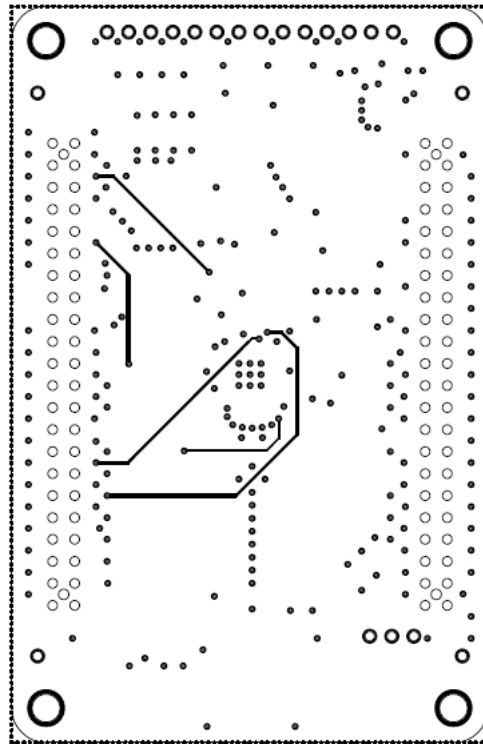


Figure 70: Inner Layer (GND)

19.3.5. Inner Layer (Supply VS)

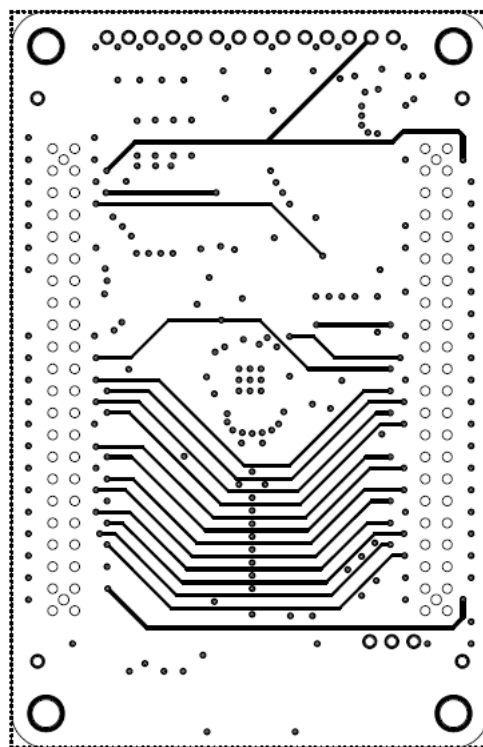
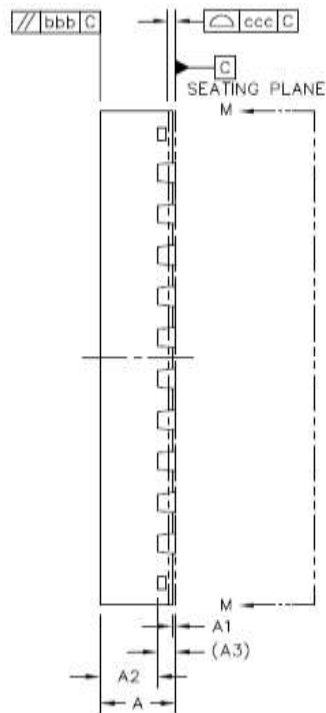


Figure 71: Inner Layer (Supply VS)



19.4. Package Dimensions



Package Dimensions				
Parameter	Ref	Min	Nom	Max
Total thickness	A	0.8	0.85	0.9
Stand off	A1	0	0.03 5	0.05
Mold thickness	A2	-	0.65	0.67
Lead frame thickness	A3	0.203 REF		
Lead width	b	0.2	0.25	0.3
Body size X	D	6 BSC		
Body size Y	E	6 BSC		
Lead pitch	e	0.5 BSC		
Exposed die pad size X	J	4.52	4.62	4.72
Exposed die pad size Y	K	4.52	4.62	4.72
Lead length	L	0.35	0.4	0.45
Package edge tolerance	aaa	0.1		
Mold flatness	bbb	0.1		
Coplanarity	ccc	0.08		
Lead offset	ddd	0.1		
Exposed pad offset	eee	0.1		

Table 93: Package Dimensions

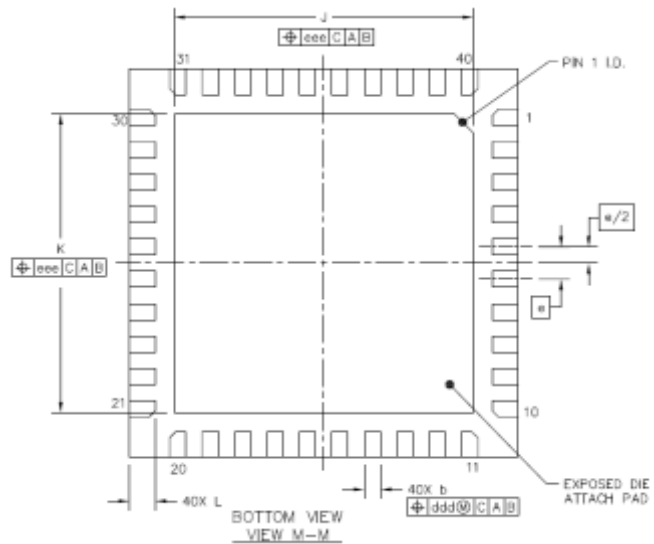
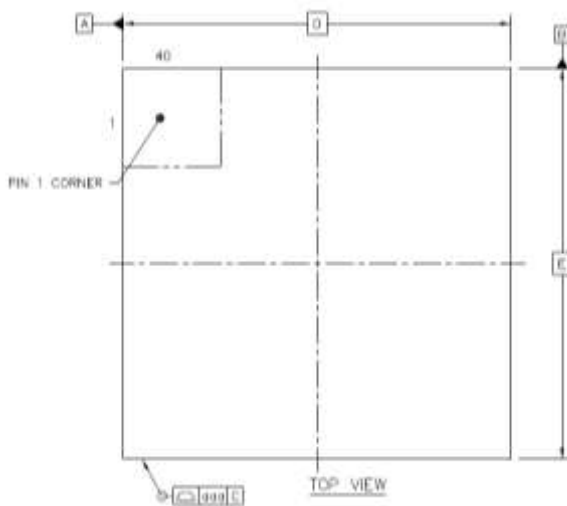


Figure 72: Package Dimensional Drawings



19.5. Package Material Information

Please refer to the associated document *"TMC43xx Package Material Information, V1.00"* for information about available package dimensions and the various tray and reel package options. This document informs you about outside dimensions per tray and/reel and the number of ICs per tray/reel. It also provides information about available packaging units and their weight, as well as box dimension and weight details for outer packaging.

The document is available for download on the TMC4361 product page at www.trinamic.com.

- i Should you require a custom-made component packaging solution or a different outer packaging solution, or have questions pertaining to the component packaging choice, please contact our customer service.

NOTE:

→ Our trays and reels are JEDEC-compliant.

19.6. Marking Details provided on Single Chip

The marking on each single chip shows:

- ① Trinamic emblem.
- ② Product code.
- ③ Date code.
- ④ Location of the copyright holder,
which is TRINAMIC in Hamburg, Germany.
- ⑤ Lot number.

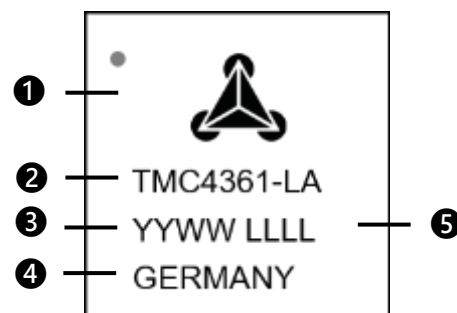


Figure 73: Marking Details on Chip¹

¹ The image provided is not an accurate rendition of the original product but only serves as illustration.



APPENDICES

20. Supplemental Directives

ESD-DEVICE INSTRUCTIONS



This product is an ESD-sensitive CMOS device. It is sensitive to electrostatic discharge.

- Provide effective grounding to protect personnel and machines.
- Ensure work is performed in a nonstatic environment.
- Use personal ESD control footwear and ESD wrist straps, if necessary.

Failure to do so can result in defects, damages and decreased reliability.

Producer Information

The producer of the product TMC4361 is TRINAMIC GmbH & Co. KG in Hamburg, Germany; hereafter referred to as TRINAMIC. TRINAMIC is the supplier; and in this function provides the product and the production documentation to its customers.

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Target User

The documentation provided here, is for programmers and engineers only, who are equipped with the necessary skills and have been trained to work with this type of product.

The **Target User** knows how to responsibly make use of this product without causing harm to himself or others, and without causing damage to systems or devices, in which the user incorporates the product.

•→Continued on next page.



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Product Documentation Details

This document ***Datasheet User Manual*** contains the ***User Information*** for the ***Target User***.

The ***Short Spec*** forms the preface of the document and is aimed at providing a general product overview. The Main Manual contains detailed product information pertaining to functions, and configuration settings. It contains all other pages of this document.

Collateral Documents & Tools

This product documentation is related and/or associated with additional tool kits, firmware and other items, as provided on the product page at: www.trinamic.com .



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23. Revision History

Document Revision History			
Version	Date	Author	Description
1.00	2014-APR-11	HS, SD	First complete version.
1.01	2014-APR-28	SD	Cross reference in chapter 9.3 corrected. Internal ramp generator units corrected.
2.00	2014-MAY-19	SD	Design changes. Change of connection scheme.
2.01	2014-MAY-23	SD	Clock frequency range updated.
2.02	2014-DEC-11	JP	Changed package suffice. Addition of "LA"
2.03	2015-MAR-02	JP	Updated temperature ratings.
3.00	2016-MAR-21	HS, SV	New document layout. Reordering of chapters. Better Navigation.
3.10	2016-JUL-04	HS, SV	New Layout, ANSI-compliant safety notices, new release variant.

Table 94: Document Revision History

